PRACTICAL NO: 2

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Roll No:11

Title: Bus Transport System

```
mysql>  create database Bus;
Query OK, 1 row affected  (0.02 sec)

mysql> use bus;
Database changed

mysql> create table route(
    ->route_no int primary key,
    ->source varchar(20),
    -> destination  varchar(34),
    ->no_of_station int
   ->);
Query OK, 0 rows affected  (0.04 sec)

mysql> desc route;
+--------------------------+----------------------+----------+--------+---------------+------------+
| Field          | Type         | Null | Key | Default | Extra |
+--------------------------+----------------------+----------+--------+---------------+------------+
| route_no       | int          | NO   | PRI | NULL    |       |
| source         | varchar(20)  | YES  |     | NULL    |       |
| destination    | varchar(34)  | YES  |     | NULL    |       |
| no_of_station | int          | YES  |     | NULL    |       |
+--------------------------+----------------------+----------+--------+---------------+------------+
4 rows in set (0.00 sec)

mysql> select * from route;
+-----------------+------------------+---------------------+--------------------------+
| route_no | source    | destination | no_of_station |
+-----------------+------------------+---------------------+--------------------------+
| 1        | mumbai    | pune        | 4             |
| 2        | kotharud  | nigdi       | 5             |
| 3        | swargate  | hadapsar    | 4             |
| 4        | karad     | kolhapur    | 6             |
| 5        | umbraj    | satara      | 7             |
| 6        | chinchwad | corporation | 5             |
+-----------------+------------------+---------------------+--------------------------+
6 rows in set (0.00 sec)

mysql> CREATE TABLE bus (
   -> bus_no INT PRIMARY KEY,
```

```
    -> capacity INT,
    -> depot_name VARCHAR(20),
    -> route_no INT,
    -> FOREIGN KEY (route_no) REFERENCES  route(route_no)
    -> );
Query OK, 0 rows affected  (0.04 sec)

mysql> desc bus;
+--------------------+----------------------+----------+---------+---------------+------------+
| Field       | Type         | Null | Key | Default | Extra |
+--------------------+----------------------+----------+---------+---------------+------------+
| bus_no      | int          | NO   | PRI | NULL    |       |
| capacity    | int          | YES  |     | NULL    |       |
| depot_name | varchar(20) | YES  |     | NULL    |       |
| route_no    | int          | YES  | MUL | NULL    |       |
+--------------------+----------------------+----------+---------+---------------+------------+
4 rows in set (0.00 sec)

mysql> select * from bus;
+-------------+-----------------+--------------------+-----------------+
| bus_no | capacity | depot_name | route_no |
+-------------+-----------------+--------------------+-----------------+
| 111     | 25        | pune        | 1         |
| 112     | 30        | kolhapur    | 4         |
| 113     | 45        | satara      | 5         |
| 114     | 20        | chinchwad  | 6         |
| 115     | 50        | swargate    | 3         |
| 116     | 20        | nigdi       | 2         |
+-------------+-----------------+--------------------+-----------------+
6 rows in set (0.00 sec)

mysql> create table driver (
    -> driver_no int primary key,
    -> driver_name  char(20),
    -> license_no  int,
    -> address char(20),
    -> d_age int,
    -> salary float,
    -> bus_no int,
    -> foreign key(bus_no) references bus(bus_no)
    -> );
Query OK, 0 rows affected  (0.04 sec)

mysql> desc driver;
+---------------------+-----------------+----------+---------+---------------+------------+
| Field        | Type      | Null | Key | Default | Extra |
+---------------------+-----------------+----------+---------+---------------+------------+
| driver_no    | int       | NO   | PRI | NULL    |       |
| driver_name | char(20) | YES  |     | NULL    |       |
| license_no  | int       | YES  |     | NULL    |       |
| address      | char(20) | YES  |     | NULL    |       |
| d_age        | int       | YES  |     | NULL    |       |
| salary       | float     | YES  |     | NULL    |       |
| bus_no       | int       | YES  | MUL | NULL    |       |
+---------------------+-----------------+----------+---------+---------------+------------+
7 rows in set (0.00 sec)
```

mysql> select * from driver;

driver_no	driver_name	license_no	address	d_age	salary	bus_no
301	rohit deshmukh	31	karad	27	22000	113
302	sana shaikh	32	chinchwad	24	27000	114
303	adnan patil	33	swargate	33	34000	115
304	imran khan	34	nigdi	39	41000	116
305	pooja jadhav	35	kolhapur	29	31000	112

5 rows in set (0.00 sec)

```
mysql> create table bus_driver  (
    -> bus_no int,
    -> driver_no  int,
    -> duty_date date,
    -> shift tinyint,  -- 1 for morning,  2 for evening
    -> primary key (bus_no, driver_no,  duty_date),
    -> foreign key (bus_no) references bus(bus_no),
    -> foreign key (driver_no)  references driver(driver_no)
    -> );
Query OK, 0 rows affected  (0.04 sec)
```

mysql> desc bus_driver;

Field	Type	Null	Key	Default	Extra
bus_no	int	NO	PRI	NULL	
driver_no	int	NO	PRI	NULL	
duty_date	date	NO	PRI	NULL	
shift	tinyint	YES		NULL	

4 rows in set (0.00 sec)

mysql> select * from bus_driver;

bus_no	driver_no	duty_date	shift
112	305	2025-09-18	1
113	301	2008-02-25	1
114	302	2008-02-13	2
115	303	2008-01-12	1
116	304	2008-01-12	2

5 rows in set (0.00 sec)

Queries:

1) Find out the drivers working in shift 1.

```
mysql> SELECT d.driver_no, d.driver_name
   -> FROM driver AS d
   -> JOIN bus_driver AS bd
   -> ON d.driver_no  = bd.driver_no
   -> WHERE bd.shift = 1;

+-----------+--------------+
| driver_no | driver_name  |
+-----------+--------------+
| 305       | pooja jadhav |
| 301       | rohit deshmukh |
| 303       | adnan patil  |
+-----------+--------------+
3 rows in set (0.00 sec)
```

2) Find out the route details on which buses of capacity 20 runs.

```
mysql> SELECT b.bus_no, b.capacity, r.route_no, r.source,
   -> r.destination,  r.no_of_station
   -> FROM bus AS b
   -> JOIN route AS r
   -> ON b.route_no = r.route_no
   -> WHERE b.capacity = 20;

+--------+----------+----------+-----------+-------------+--------------+
| bus_no | capacity | route_no | source    | destination | no_of_station |
+--------+----------+----------+-----------+-------------+--------------+
| 114    | 20       | 6        | chinchwad | corporation | 5            |
| 116    | 20       | 2        | kotharud  | nigdi       | 5            |
+--------+----------+----------+-----------+-------------+--------------+
2 rows in set (0.00 sec)
```

3) Find the names and their license no. of driver working on 12-01-2008 in both the shifts.

```
mysql> SELECT d.driver_name, d.license_no
   -> FROM bus_driver AS bd
   -> JOIN driver AS d
   -> ON bd.driver_no  = d.driver_no
   -> WHERE bd.duty_date = '2008-01-12';

+--------------+------------+
| driver_name  | license_no |
+--------------+------------+
| adnan patil  | 33         |
| imran khan   | 34         |
+--------------+------------+
2 rows in set (0.00 sec)
```

4) Delete all the routes on which total stations are less than 3.

```
mysql> DELETE FROM route
   -> WHERE no_of_station < 3;
```

Query OK, 0 rows affected (0.01 sec)
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

5) Find out the number of buses running from 'chinchwad' to 'corporation'.

```
mysql> SELECT COUNT(*) AS num_buses
    -> FROM bus
    -> JOIN route ON bus.route_no = route.route_no
    -> WHERE route.source = 'chinchwad'
    -> AND route.destination = 'corporation';

+-----------+
| num_buses |
+-----------+
| 1         |
+-----------+
1 row in set (0.01 sec)
```

6) Update the salary of driver by 1000 if his age >35.

```
mysql> UPDATE driver
    -> SET salary = salary + 1000
    -> WHERE d_age > 35;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from driver;
+-----------+--------------+-----------+-----------+-------+--------+--------+
| driver_no | driver_name  | license_no | address   | d_age | salar  | bus_no |
+-----------+--------------+-----------+-----------+-------+--------+--------+
| 301       | rohit deshmukh| 31        | karad     | 27    | 22000  | 113    |
| 302       | sana shaikh  | 32        | chinchwad | 24    | 27000  | 114    |
| 303       | adnan patil  | 33        | swargate  | 33    | 34000  | 115    |
| 304       | imran khan   | 34        | nigdi     | 39    | 42000  | 116    |
| 305       | pooja jadhav | 35        | kolhapur  | 29    | 31000  | 112    |
+-----------+--------------+-----------+-----------+-------+--------+--------+
5 rows in set (0.00 sec)
```

7) List the bus numbers which are running from 'swargate'-'hadapsar' having bus capacity50.

```
mysql> SELECT b.bus_no
    -> FROM bus AS b
    -> JOIN route AS r
    -> ON b.route_no = r.route_no
    -> WHERE r.source = 'swargate'
    -> AND r.destination = 'hadapsar'
    -> AND b.capacity = 50;

+--------+
| bus_no |
+--------+
| 115    |
+--------+
1 row in set (0.00 sec)
```

Stored Functions:

1) Write a function to find out the name of the driver having maximum salary.

```
mysql> DELIMITER $$
mysql> CREATE FUNCTION get_max_salary_driver()
    -> RETURNS VARCHAR(100)
    -> DETERMINISTIC
    -> BEGIN
    -> DECLARE driver_name_var  VARCHAR(100);
    -> SELECT driver_name
    -> INTO driver_name_var
    -> FROM driver
    -> WHERE salary = (SELECT MAX(salary) FROM driver)
    -> LIMIT 1;
    -> RETURN driver_name_var;
    -> END$$
Query OK, 0 rows affected  (0.01 sec)

mysql> DELIMITER ;
mysql> SELECT get_max_salary_driver();
+-----------------------------------------+
| get_max_salary_driver() |
+-----------------------------------------+
| imran khan                |
+-----------------------------------------+
1 row in set (0.01 sec)
```

2) Write a function to accept the bus_no and date and print its allotted driver.

```
mysql> DELIMITER $$

mysql> CREATE PROCEDURE get_assigned_driver (
    -> IN in_bus_no  INT,
    -> IN in_duty_date DATE
    -> )
    -> BEGIN
    -> SELECT d.driver_name
    -> FROM driver AS d
    -> JOIN bus_driver AS bd
    -> ON d.driver_no  = bd.driver_no
    -> WHERE bd.bus_no = in_bus_no
    -> AND bd.duty_date = in_duty_date;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>  DELIMITER  ;

mysql>  CALL get_assigned_driver(116,  '2008-01-12');
+--------------------------+
| driver_name    |
+--------------------------+
| imran khan     |
+--------------------------+
```

1 row in set (0.01 sec)
Query OK, 0 rows affected (0.01 sec)

STORED PROCEDURE:

1) Write a procedure to list the drivers working on a particular date shift wise. (Accept the date as an input parameter).

```
mysql> DELIMITER $$

mysql> CREATE PROCEDURE list_drivers_by_date (
   -> IN in_duty_date DATE
   -> )
   -> BEGIN
   -> SELECT bd.shift,
   -> d.driver_no,
   -> d.driver_name,
   -> d.license_no
   -> FROM bus_driver AS bd
   -> JOIN driver AS d
   -> ON bd.driver_no  = d.driver_no
   -> WHERE bd.duty_date = in_duty_date
   -> ORDER BY bd.shift,  d.driver_no;
   -> END$$
Query OK, 0 rows affected  (0.00 sec)

mysql> DELIMITER ;
mysql> CALL list_drivers_by_date('2008-01-12');
+------------+------------------+-------------------------+--------------------+
| shift | driver_no | driver_name    | license_no |
+------------+------------------+-------------------------+--------------------+
| 1      | 303        | adnan patil    | 33          |
| 2      | 304        | imran khan     | 34          |
+------------+------------------+-------------------------+--------------------+
2 rows in set (0.00 sec)

Query OK, 0 rows affected  (0.01 sec)
```

2) List the bus_no and drivers allocated to the buses which are running from 'kotharud' to 'nigdi' on date.

```
mysql> DELIMITER $$

mysql> CREATE PROCEDURE get_kotharud_to_nigdi_drivers  (
   -> IN in_duty_date DATE
   -> )
   -> BEGIN
   -> SELECT bd.bus_no,
   -> d.driver_name
   -> FROM bus_driver AS bd
   -> JOIN bus AS b
   -> ON bd.bus_no = b.bus_no
   -> JOIN driver AS d
   -> ON bd.driver_no  = d.driver_no
   -> JOIN route AS r
      -> ON b.route_no = r.route_no
```

```
    -> WHERE r.source = 'kotharud'
    -> AND r.destination  = 'nigdi'
    -> AND bd.duty_date = in_duty_date
    -> ORDER BY bd.bus_no;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;

mysql> CALL get_kotharud_to_nigdi_drivers('2008-01-12');
+--------------+--------------------------+
| bus_no | driver_name    |
+--------------+--------------------------+
| 116    | imran khan     |
+--------------+--------------------------+
1 row in set (0.00 sec)
```

Cursor:

1) Write a procedure using cursor to display the details of a driver, (accept driver name as input parameter).

```
mysql> DELIMITER $$

mysql> CREATE PROCEDURE get_driver_details_by_name(
   -> IN driver_name_input  VARCHAR(100)
   -> )

   -> BEGIN
   -> DECLARE done INT DEFAULT FALSE;
   -> DECLARE driver_no_var  INT;
   -> DECLARE license_no_var  INT;
   -> DECLARE address_var VARCHAR(255);
   -> DECLARE d_age_var INT;
   -> DECLARE salary_var  DECIMAL(10,2);
   -> DECLARE bus_no_var  INT;
   -> DECLARE driver_cursor  CURSOR FOR
   -> SELECT driver_no,  license_no,  address, d_age, salary, bus_no
   -> FROM driver
   -> WHERE LOWER(TRIM(driver_name))  =
   -> LOWER(TRIM(driver_name_input));

   -> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
   -> OPEN driver_cursor;
   -> read_loop: LOOP
   -> FETCH driver_cursor  INTO driver_no_var,  license_no_var,
   -> address_var, d_age_var, salary_var,  bus_no_var;
   -> IF done THEN
   -> LEAVE read_loop;
   -> END IF;

   -> SELECT driver_no_var  AS "Driver  No",
   -> driver_name_input AS "Driver  Name",
   -> license_no_var AS "License  No",
   -> address_var AS "Address",
   -> d_age_var AS "Age",
   -> salary_var AS "Salary",
   -> bus_no_var AS "Bus No";
   -> END LOOP;
   -> CLOSE driver_cursor;
   -> END$$

Query OK, 0 rows affected (0.01 sec)

mysql>  DELIMITER  ;

mysql> CALL get_driver_details_by_name('sana  shaikh');
+------------------+-----------------------+--------------------+-----------------+----------+-------------+-------------+
| Driver No | Driver Name   | License No | Address   | Age   | Salary | Bus No |
+------------------+-----------------------+--------------------+-----------------+----------+-------------+-------------+
| 302         | sana shaikh   | 32           | chinchwad| 24    | 27000   | 114      |
+------------------+-----------------------+--------------------+-----------------+----------+-------------+-------------+
1 row in set (0.00 sec)
```

2) Write a cursor to display the details the details of the buses that run on route_no=1 and route_no=2.
(Use two different cursors for route_no=1 and route_no=2).

```
mysql> DELIMITER $$


mysql> CREATE PROCEDURE display_buses_route_1_and_2()
   -> BEGIN

   -> DECLARE bus_no_1 INT;
   -> DECLARE capacity_1  INT;
   -> DECLARE depot_1 VARCHAR(20);
   -> DECLARE bus_no_2 INT;
   -> DECLARE capacity_2  INT;
   -> DECLARE depot_2 VARCHAR(20);

   -> BEGIN
   -> DECLARE done1 INT DEFAULT FALSE;

   -> DECLARE cursor_route1  CURSOR FOR
   -> SELECT bus_no, capacity,  depot_name
   -> FROM bus
   -> WHERE route_no = 1;

   -> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done1 = TRUE;
   -> OPEN cursor_route1;

   -> route1_loop:  LOOP
   -> FETCH cursor_route1  INTO bus_no_1, capacity_1,  depot_1;
   -> IF done1 THEN
   -> LEAVE route1_loop;
   -> END IF;

   -> SELECT bus_no_1 AS "Bus No",
   -> capacity_1 AS "Capacity",
   -> depot_1 AS "Depot",
   -> 1 AS "Route  No";
   -> END LOOP;

   -> CLOSE cursor_route1;
   -> END;

   -> BEGIN
   -> DECLARE done2 INT DEFAULT FALSE;
   -> DECLARE cursor_route2  CURSOR FOR
   -> SELECT bus_no, capacity,  depot_name
   -> FROM bus
   -> WHERE route_no = 2;
   -> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done2 = TRUE;

   -> OPEN cursor_route2;
   -> route2_loop:  LOOP
   -> FETCH cursor_route2  INTO bus_no_2, capacity_2,  depot_2;
   -> IF done2 THEN
```

```
    -> LEAVE route2_loop;
    -> END IF;

    -> SELECT bus_no_2 AS "Bus No",
    -> capacity_2 AS "Capacity",
    -> depot_2 AS "Depot",
    -> 2 AS "Route  No";
    -> END LOOP;

    -> CLOSE cursor_route2;
    -> END;
    -> END$$
```

Query OK, 0 rows affected (0.01 sec)

```
mysql>  DELIMITER  ;
```

```
mysql> CALL display_buses_route_1_and_2();
+--------------+-----------------+------------+-------------------+
| Bus No | Capacity | Depot | Route No  |
+--------------+-----------------+------------+-------------------+
| 111     | 25        | pune  | 1          |
+--------------+-----------------+------------+-------------------+
1 row in set (0.00 sec)

+--------------+-----------------+------------+-------------------+
| Bus No | Capacity | Depot | Route No  |
+--------------+-----------------+------------+-------------------+
| 116     | 20        | nigdi | 2          |
+--------------+-----------------+------------+-------------------+
1 row in set (0.00 sec)
```

Trigger:

1) Define a trigger after insert or update the record of driver if the age is between 18 and 50 give the message "valid entry" otherwise give appropriate message.

FOR INSERT:

```
mysql> DELIMITER $$

mysql> CREATE TRIGGER  validate_driver_age_insert
   -> AFTER INSERT ON driver
   -> FOR EACH ROW
   -> BEGIN
   -> IF NEW.d_age NOT BETWEEN  18 AND 50 THEN
   -> SIGNAL SQLSTATE '45000'
   -> SET MESSAGE_TEXT  =
   -> 'Invalid entry: Driver age must be between 18 and 50.';
   -> END IF;
   -> END$$

Query OK, 0 rows affected  (0.01 sec)

mysql>  DELIMITER  ;
mysql> insert into driver
   -> values(306,'ayesha  more',36,'pune',16,28000,111);
ERROR 1644 (45000): Invalid entry: Driver age must be between 18 and 50.
```

FOR UPDATE:

```
mysql> DELIMITER $$

mysql> CREATE TRIGGER  validate_driver_age_update
   -> AFTER UPDATE ON driver
   -> FOR EACH ROW
   -> BEGIN
   -> IF NEW.d_age NOT BETWEEN  18 AND 50 THEN
   -> SIGNAL SQLSTATE '45000'
   -> SET MESSAGE_TEXT  =
   -> 'Invalid entry: Driver age must be between 18 and 50.';
   -> END IF;
   -> END$$

Query OK, 0 rows affected (0.01 sec)

mysql>  DELIMITER  ;

mysql>  update driver set d_age=51 where driver_no=305;
ERROR 1644 (45000): Invalid entry: Driver age must be between 18 and 50.
```

2) Define a trigger after delete the record of bus having

capacity <10.display the message accordingly.

```
mysql>  CREATE TABLE  bus_delete_log
   -> (
   -> bus_no INT,
```

```
    -> capacity INT,
    -> message VARCHAR(100)
    -> );
Query OK, 0 rows affected  (0.01 sec)

mysql> DELIMITER $$

mysql> CREATE TRIGGER  after_bus_delete
    -> AFTER DELETE ON bus
    -> FOR EACH ROW
    -> BEGIN
    -> IF OLD.capacity < 10 THEN
    -> INSERT INTO bus_delete_log  (bus_no, capacity, message)
    -> VALUES (OLD.bus_no, OLD.capacity,
    -> 'Invalid  delete: capacity < 10');
    -> ELSE
    -> INSERT INTO bus_delete_log  (bus_no, capacity, message)
    -> VALUES (OLD.bus_no, OLD.capacity, 'Valid delete');
    -> END IF;
    -> END$$

Query OK, 0 rows affected (0.01 sec)

mysql>  DELIMITER  ;

mysql> insert into bus values(117,8,'test depot',1);
Query OK, 1 row affected  (0.01 sec)

mysql> delete from bus where bus_no=117;
Query OK, 1 row affected  (0.01 sec)

mysql> select * from bus_delete_log;
+--------------+-----------------+-------------------------------------------------+
| bus_no | capacity | message                                          |
+--------------+-----------------+-------------------------------------------------+
| 117    | 8        | Invalid delete: capacity < 10|
+--------------+-----------------+-------------------------------------------------+
1 row in set (0.00 sec)
```